



Introduction to

Internet of Things

Assignment-Week 12

TYPE OF QUESTION: MCQ

Number of questions: 15

Total marks: 15 X 1 = 15

Fill in the blank.

The two broad types of data analysis are _____ and _____.

a. Qualitative and Quantitative

b. Repetitive and Quantitative

c. Repetitive and Qualitative

d. All of these

Correct Answer: a. Qualitative and Quantitative

Detailed Solution: Qualitative and quantitative are the two types of data analysis. (Please refer Lecture 56@2:33)

QUESTION 2:

Which of the following is/are the basic principle(s) of qualitative analysis?

a. Notice things

b. Think about things

c. Collect things

d. All of these

Correct Answer: d. All of these

Detailed Solution: The principles include noticing, thinking, and collecting. (Please refer Lecture 56@4:12)

QUESTION 3:



NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur



Fill in the blank.

To perform an ANOVA, we must have a ____ response variable and at least one ____ factor.

- a. Discrete, categorical
- b. Continuous, quantitative
- c. Discrete, quantitative

d. Continuous, categorical

Correct Answer: d. Continuous, categorical

Detailed Solution: ANOVA requires a continuous response and at least one categorical factor. (Please refer Lecture 56@9:24)

QUESTION 4:

Which of the following is/are application(s) of IoT in agriculture?

- a. Automated irrigation system
- b. Automated weeding system
- c. Water level monitoring

d. All of these

Correct Answer: d. All of these

Detailed Solution: All listed options are agriculture IoT applications. (Please refer Lecture 57@2:05)

QUESTION 5:

The two most directly relevant sensors used in agriculture (as discussed) are ____.

- a. Soil moisture and proximity sensor
- b. Soil moisture and water level sensor**
- c. ECG sensor and water level sensor
- d. All of these

Correct Answer: b. Soil moisture and water level sensor



Detailed Solution: Soil moisture and water level sensors are commonly used in agriculture monitoring. (Please refer Lecture 57@13:11)

QUESTION 6:

The deployed IoT gateway of AgriSens communicates with the remote server using:

- a. WiFi
- b. Bluetooth
- c. Internet Phone System

d. GPRS

Correct Answer: d. GPRS

Detailed Solution: AgriSens gateway communicates to remote server using GPRS. (Please refer Lecture 57@9:14)

QUESTION 7:

Select the correct order of component layers in IoT healthcare.

- a. Sensing layer, cloud platform layer, aggregated layer, processing layer
- b. Sensing layer, aggregated layer, processing layer, cloud platform layer**
- c. Aggregated layer, sensing layer, processing layer, cloud platform layer
- d. Sensing layer, processing layer, aggregated layer, cloud platform layer

Correct Answer: b. Sensing layer, aggregated layer, processing layer, cloud platform layer

Detailed Solution: Data flows from sensing → aggregation → processing → cloud platform. (Please refer Lecture 58@8:25)

QUESTION 8:

In AmbuSens, which communication protocol is used?

- a. IEEE 802.15.4



b. 6LoWPAN

c. IEEE 802.15.1

d. IEEE 802.15.2

Correct Answer: c. IEEE 802.15.1

Detailed Solution: AmbuSens uses Bluetooth, i.e., IEEE 802.15.1. (Please refer Lecture 58@22:09)

QUESTION 9:

State True or False.

Wireless IoT-driven remote healthcare provisioning brings healthcare to patients rather than bringing patients to healthcare.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Remote IoT healthcare aims to bring healthcare to the patient. (Please refer Lecture 58@11:40)

QUESTION 10:

Which handheld device sensor pair is used for activity monitoring (as discussed)?

a. EEG and GPS

b. Accelerometer and EEG

c. Accelerometer and GPS

d. All of these

Correct Answer: c. Accelerometer and GPS

Detailed Solution: Accelerometer and GPS are used for activity monitoring in handheld devices. (Please refer Lecture 59@15:29)



QUESTION 11:

Which of the following is a property of in-place data analysis for activity monitoring?

- a. Power intensive
- b. On the device
- c. No need of network

d. All of these

Correct Answer: d. All of these

Detailed Solution: In-place analysis is on-device, power intensive, and can work without network. (Please refer Lecture 59@20:24)

QUESTION 12:

State True or False.

Deep learning-based analysis cannot be performed on videos.

- a. True

b. False

Correct Answer: b. False

Detailed Solution: Deep learning can be applied to sensor data, images, and videos. (Please refer Lecture 59@19:34)

QUESTION 13:

Which of the following are inbuilt sensors present in high-end smartphones (as discussed)?

- a. ECG and EEG
- b. Accelerometer, proximity sensor, and EEG

c. Accelerometer, Compass, and Gyroscope

- d. Pressure sensor and NPK sensor

Correct Answer: c. Accelerometer, Compass, and Gyroscope



Detailed Solution: High-end smartphones commonly have accelerometer, compass, and gyroscope. (Please refer Lecture 60@2:28)

QUESTION 14:

Fill in the blank.

Processing the handheld activity device data using artificial intelligence can be used for _____.

a. Fall detection

b. Heart rate detection

c. Vehicle detection

d. All of these

Correct Answer: a. Fall detection

Detailed Solution: AI processing of activity data can be used to detect sudden falls. (Please refer Lecture 60@11:56)

QUESTION 15:

State True or False.

Continuous monitoring of a person's activity can help observe behavior and identify repetitive patterns.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Continuous monitoring enables observation of behavior and repetitive activity patterns. (Please refer Lecture 59@14:33)



NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur



*****END*****